Section: A QUIZ - 3 Time Allowed: 20 mins

Name Solution Roll No.

1. Which operation increases the number of feature maps without changing spatial size?
A. Pooling B. Convolution with multiple filters C. Strided convolution D. Padding

2. In CNNs, deeper layers primarily learn:
A. Edges and textures B. Color histograms C. Task-specific abstract patterns D. Kernel sizes

3. What does max pooling achieve in a CNN?
A. Increases spatial resolution B. Reduces spatial dimensions C. Adds more parameters D. Performs normalization

4. Early stopping helps most with:
A. Underfitting B. GPU memory usage C. Overfitting D. Batch normalization

5. The primary benefit of data augmentation in CNN training is:
A. Faster training B. Reduced dataset size C. Model compression D. Improved generalization

6. In PyTorch, which module typically stores trainable weights?
A. torch.nn.Module B. torch.utils.data.Dataset C. torch.autograd D. torch.optim

7. Which layer is primarily responsible for extracting spatial features in CNNs?
A. Fully connected layer B. Convolution layer C. Recurrent layer D. Softmax layer

8. Which PyTorch class is used to define a neural network model?
A. torch.nn.Module B. torch.nn.Trainer C. torch.model.Network D. torch.autograd.Model

9. What is the main purpose of batch normalization?
A. Reduce overfitting B. Stabilize activations and speed training C. Increase dropout D. Add noise to inputs

10. A 3×3 filter with padding=1 keeps the spatial size same because:
A. Padding adds learnable parameters B. Padding increases batch size
C. Padding preserves boundary information D. Filter size becomes irrelevant

11. Which of the following best describes what a convolutional filter does in a CNN?
A. It reduces the size of the dataset by random sampling B. It detects specific patterns such as edges or textures in the input C. It converts the input image to grayscale D. It normalizes pixel values to a fixed range

12. What is the main benefit of using GPU acceleration?
A. Higher accuracy B. Faster parallel computation C. Lower memory usage D. Automatic tuning

13. Which layer is typically used at the end of a classification network?
A. Conv2D B. MaxPool C. Fully Connected D. RNN

14. In PyTorch, which function switches the model to evaluation mode?
A. model.activate() B. model.eval() C. model.test() D. model.freeze()

15. A 3×3 convolution kernel contains how many weights per channel?
A. 6 B. 9 C. 3 D. 12

16. Which PyTorch object loads data in batches?
A. torch.nn.Dataset B. torch.utils.data.DataLoader C. torch.loader.Batch D. torch.batch

17. What problem does the ResNet skip connection primarily solve?
A. Overfitting B. Vanishing gradients in deep networks C. Slow training due to large datasets
D. High GPU memory consumption

18. In PyTorch, which operation triggers gradient computation during backpropagation?
A. Calling model.train() B. Setting requires_grad=False C. Running loss.backward() D. Calling optimizer.zero_grad()

19. Which technique is most commonly used to reduce overfitting in neural networks?
A. Increasing batch size B. Adding dropout layers C. Removing hidden layers D. Using only training data for evaluation

20. L2 regularization encourages:
A. Sparsity B. Smaller weight magnitudes C. More dropout D. More pooling